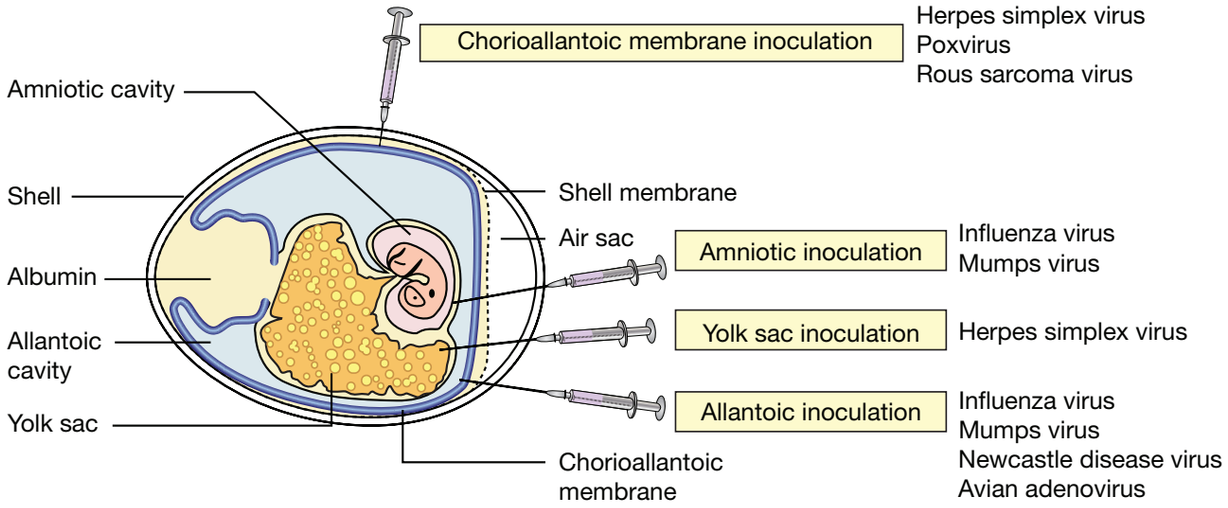


Figure 10.3: Cultivation of virus in Embryonated Egg

iv. Cell Culture

Tissues are dissociated into the component cells by the **action of enzymes** (trypsin) or by mechanical process and are suspended in a growth medium (amino acids, vitamins, salts, glucose) **supplemented** with fetal calf serum of antibiotics and **indicator** (Phenol red). This media is dispensed in bottles, tubes or petridishes. The cells adhere to the glass surface and on incubation divides to form a confluent monolayer sheet of cells covering the surface within about a week. The cell culture is classified into three types.

a Primary cell cultures

In this culture, **normal cells** are taken from the body and cultured. They are capable of only limited growth in culture. Example: Monkey kidney, Human embryonic kidney, Chick embryo cell culture.

b. Diploid cell strains

These are cells of a **single type** that retain the original **diploid chromosome number** and serotype during serial sub cultivation for limited number of times. Example: Human fibroblast.

c. Continuous cell lines

These are **single type, derived from cancer cells** that are capable of continuous serial cultivation. Example: Cells derived from cancers, such as Hela, Hep-2 and KB cell lines.

Detection of virus Growth in Cell Cultures

Virus growth in cell cultures can be detected by the following methods.

1. Cytopathic effect

Many viruses cause morphological changes in cultured cells in which they grow. These changes are known as **cytopathic effects (CPE) and** the viruses causing CPE are called as **'cytopathogenic viruses'.**

2. Metabolic inhibition

In normal cell cultures, the medium **turns acidic** due to cellular metabolism. This can be made out by the colour of the indicator (Phenol red) incorporated in the medium.

3. Haemadsorption

When **haemagglutinating viruses** grow in cell cultures, their presence can be identified by the addition of guinea pig

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